PUNYASHLOK AHILYADEVI HOLKAR SOLAPUR UNIVERSITY, SOLAPUR

B.SC.[ECS] - II Sem III Practical Examination, October 2026

VSC1: Based on DSC1-3 and DSC1-4

Date & Time: Max Marks: 30

Instructions:

I) Attempt any **one** question from each section.

II) Each question carries **10 marks.**

III) The **Journal** and **Viva-Voce** carry **5 marks.**

Section-I: Based on DSC1-3 (Data Structures using C++)

Q.1 Write a program to implement binary search tree with insert, postorder, search operations.

Q.2 Write a program to evaluate a postfix expression using stack.

Section-II: Based on DSC1-4 (Database Management Systems)

Q.3 Create a table by applying proper constraints, insert at least 5 records and solve the following queries:

Field Name	Datatype	Size	Description
sno	Varchar2	6	Primary Key
sname	Varchar2	20	Account Holders name
address	Varchar2	30	Residential address
address1	Varchar2	30	Permanent address
pincode	number	6	Pin code
dob	Date		Date of birth
state	char	20	State
doj	date		Date of joining must be greater than Date of birth
department	Char	20	department name
salary	number	9,2	Salary of salesman

a) List of all salesmen whose address starts with 'N'.

b) List of salesmen whose Department is not Computer and HR.

c) List of salesmen whose Mobile_no is 9423376798 or 9763658485.

d) List of salesmen who joined in the month of June.

e) List of salesmen whose salary is > 5000 and < 10000.

Q.4 Create a table by applying proper constraints, insert at least 5 records and solve the following queries:

A] Table Structure of Account_Master

Field Name	Datatype	Size	Description
Accent	Varchar2	6	Primary Key
Acc_name	Varchar2	6	Account holder's name
Gender	Char	1	Either M or F
Op_Date	Date		The date on which the Account is opened
Chq_fac	Char	1	Cheque facility status set to default 'N'
Cl_Date	Date		

B] Table Structure of Account_Transaction

Field Name	Datatype	Size	Description
Accent	Varchar2	6	Foreign Key
Tr_amount	Number	7,2	Transaction Amount
Tr_date	Date	1	Transaction Date
Chq_no	Varchar2	10	Cheque Number
Tr_type	Char	1	Receipt or Withdrawal

a) Display the accounts of those who have deposited.

b) Display the details of account holders who opened an account in 2012.

c) Display the no of female account holders having cheque facility.

d) Display the details of 2^{nd} min balance of the account holder.

e) Display the account_no, account_nm, transaction_amount, and transaction_type of all account holders.